YUKEN INDIA Ltd.

Summer Internship — PCB Design Project Report

Industrial Analog Condition Monitoring Board

An Analog PCB Design

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Design Tool	KiCad 10
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1. Abstract

This report documents the design and development of the Industrial Analog Condition Monitoring Board, a printed circuit board (PCB) created as the culminating project of the Analog Electronics phase of the internship at Yuken India Ltd.

The board monitors three independent physical parameters — temperature, pressure and ambient light — using purely analog circuitry. All decision-making is performed without a microcontroller, demonstrating the application of operational amplifiers, comparators, and transistor-based logic to real-world industrial monitoring scenarios.

The design was completed in KiCad 10, progressing through schematic capture, PCB layout, design rule verification (DRC/ERC), and Gerber file generation. The board achieved 0 errors and 0 warnings on final verification.

2. Project Objectives

The project was designed to consolidate and demonstrate the following competencies developed during the Analog Electronics internship weeks:

- Apply operational amplifier theory — non-inverting amplifier gain configuration
- Implement comparator-based threshold detection for multi-level classification
- Design Schmitt trigger circuits with controlled hysteresis
- Perform signal conditioning — low-pass filtering, amplification, and reference generation
- Construct analog decision logic using discrete transistors (no digital ICs)
- Execute a complete PCB design workflow: schematic → layout → DRC → Gerber output
- Work with mixed-technology PCB — SMD passives/ICs and through-hole connectors/transistors
- Produce a manufacturable PCB with organised layout, clean routing, and a ground plane

3. System Overview

The board contains three independent monitoring channels, each processing a sensor input through signal conditioning and comparator logic to drive a set of colour-coded LED indicators. A common power supply section generates a regulated 5 V rail from a 12 V DC input.

Channel	Sensor	Indicators	Key Technique
Temperature	LM35 (external)	Green / Yellow / Red LEDs	Op-Amp amplification + dual comparators + transistor AND logic

Channel	Sensor	Indicators	Key Technique
Pressure	Potentiometer / analog signal (external)	Green / Yellow / Red LEDs	Resistor ladder references + dual comparators + transistor AND logic
Light	LDR (external)	Green / Red LEDs	Voltage divider + Schmitt trigger hysteresis

The design intentionally excludes microcontrollers, FPGAs, and digital logic ICs. All threshold decisions, zone detection, and output control are performed by analog components.

4. Power Supply Section

4.1 Input and Protection

The board accepts a 12 V DC input through a 2-pin terminal block (J1). A 1N5819 Schottky diode (D1) provides reverse polarity protection. The 1N5819 was selected for its low forward voltage drop (~0.3 V), minimizing power dissipation while protecting the downstream circuitry.

4.2 Voltage Regulation

An LM7805 linear voltage regulator produces a regulated 5 V output. Decoupling capacitors on both the input and output sides of the regulator ensure stability and ripple suppression:

Location	Capacitor	Purpose
Input side	100 nF ceramic	High-frequency noise bypass
Input side	10 µF electrolytic	Bulk decoupling, ripple reduction
Output side	100 nF ceramic	High-frequency noise bypass
Output side	10 µF electrolytic	Load transient response

A power indicator LED (D2) with a 680 Ω current-limiting resistor (R1) confirms 5 V rail presence. The LED operating current is approximately 4.4 mA.

5. Temperature Monitoring Channel

5.1 Sensor Interface

An external LM35 precision temperature sensor connects via a 3-pin terminal block (J2: +5 V, signal, GND). The LM35 produces a linear output of 10 mV/°C, requiring amplification to span a useful comparison range.

5.2 Amplifier Stage — U2A (LM358)

The first op-amp section (U2A) is configured as a non-inverting amplifier:

Parameter	Value	Notes
Configuration	Non-inverting amplifier	—
Feedback resistor Rf	40 kΩ	R2
Input resistor Rin	10 kΩ	R3
Gain Av	5 V/V	Av = 1 + Rf/Rin = 1 + 40k/10k
Output at 40°C	2 V	40°C × 10 mV/°C × 5 = 2 V
Output at 60°C	3 V	60°C × 10 mV/°C × 5 = 3 V

5.3 Low-Pass Filter

A first-order RC low-pass filter (R = 10 kΩ, C = 1 μF) with a cutoff frequency of approximately 16 Hz is placed before the comparator inputs to suppress high-frequency noise from the sensor signal and power supply.

5.4 Threshold Detection — U3 (LM339)

Two comparator sections of the LM339 quad comparator IC (U3) implement dual-threshold detection:

Comparator	Reference Voltage	Temperature Equivalent	Output Net	Active When
U3A	2 V	40°C	TEMP_GT_40_N	Temperature > 40°C
U3B	3 V	60°C	TEMP_GT_60_N	Temperature > 60°C

5.5 LED Output Logic — Q1, Q2 (BC547)

Three zones are indicated with colour-coded LEDs. The green and red LEDs respond directly to the comparator outputs. The yellow LED (warning zone: 40–60°C) requires a logical AND of the two comparator outputs, implemented with two BC547 NPN transistors:

- Q1 (driver): switched ON by TEMP_GT_40_N — enables current path for yellow LED
- Q2 (inhibit): switched ON by TEMP_GT_60_N — pulls Q1 base LOW, disabling yellow LED

Condition	Green LED	Yellow LED	Red LED
T < 40°C	ON	OFF	OFF
40°C ≤ T < 60°C	OFF	ON	OFF
T ≥ 60°C	OFF	OFF	ON

6. Pressure Monitoring Channel

6.1 Sensor Interface

An external analog signal (0–5 V range) connects via J4, representing industrial pressure. A potentiometer is used for simulation and testing. No amplification is required since the input already spans the full 5 V reference range.

6.2 Signal Conditioning

A first-order RC low-pass filter (R = 10 kΩ, C = 1 μF) with a cutoff frequency of approximately 16 Hz smooths the input before it reaches the comparator network, preventing transient noise from causing false threshold crossings.

6.3 Reference Voltage Ladder

Three equal 10 kΩ resistors (R12, R13, R14) form a voltage divider across the 5 V rail, generating two reference points:

Reference	Voltage	Represents	Used By
Lower threshold	1.67 V (5/3)	Lower third of pressure range	Comparator U3C
Upper threshold	3.33 V (10/3)	Upper third of pressure range	Comparator U3D

6.4 LED Output Logic — Q3, Q4 (BC547)

Identical transistor logic to the temperature channel is used for yellow LED zone detection (1.67 V – 3.33 V), with Q3 as the driver and Q4 as the inhibit transistor.

Condition	Green LED	Yellow LED	Red LED
Signal < 1.67 V	ON	OFF	OFF
1.67 V ≤ Signal < 3.33 V	OFF	ON	OFF
Signal ≥ 3.33 V	OFF	OFF	ON

7. Light Monitoring Channel

7.1 Sensor Interface and Voltage Divider

An external LDR (Light Dependent Resistor) connects via J3. On the PCB, a 10 kΩ fixed resistor forms a voltage divider with the LDR between 5 V and GND. The mid-point voltage (LIGHT_SIGNAL) varies inversely with illumination — bright conditions produce a higher voltage, dark conditions a lower voltage.

7.2 Schmitt Trigger — U2B (LM358)

The second op-amp section (U2B) is configured as an inverting Schmitt trigger to provide hysteresis, preventing LED flicker at the switching threshold:

Parameter	Value	Notes
Reference voltage	2.5 V	10k/10k divider from 5 V rail
Positive feedback resistor	100 kΩ	R20 — sets hysteresis band
Output net	LIGHT_CONDITION	HIGH = bright, LOW = dark

The positive feedback through the 100 kΩ resistor creates a hysteresis band around the 2.5 V reference, ensuring the output switches cleanly between states without oscillation near the threshold.

7.3 LED Output

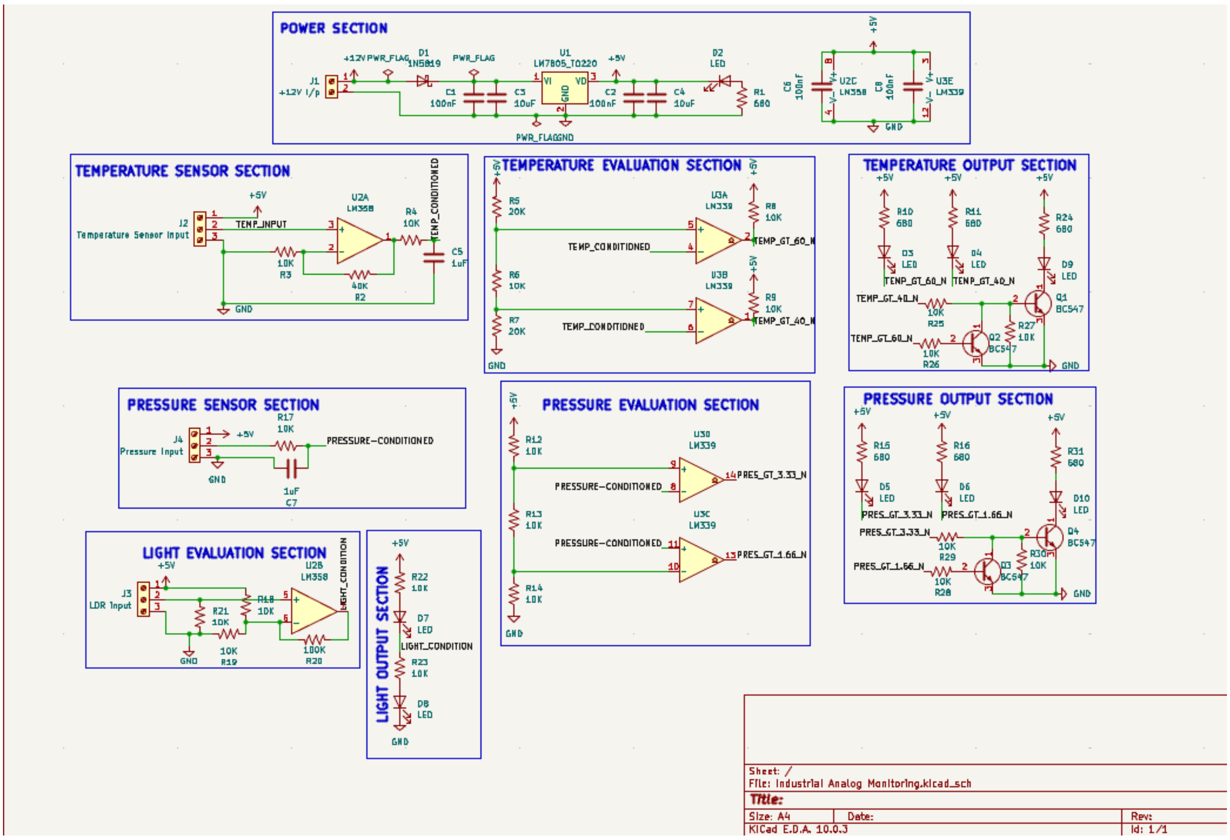
- Green LED: illuminated when LIGHT_CONDITION is HIGH (bright environment)
- Red LED: illuminated when LIGHT_CONDITION is LOW (dark environment)

8. Bill of Materials — Key Components

Reference	Component	Value / Type	Package	Quantity	Function
U1	LM7805	5 V regulator	TO-220	1	Power regulation
U2	LM358	Dual op-amp	SOIC-8	1	Amplifier + Schmitt trigger
U3	LM339	Quad comparator	SOIC-14	1	All threshold detection
D1	1N5819	Schottky diode	DO-41	1	Reverse polarity protection

Reference	Component	Value / Type	Package	Quantity	Function
Q1–Q4	BC547	NPN transistor	TO-92	4	Yellow LED AND logic
D2–D10	LED	Various colours	1206 SMD	9	Status indicators
R (various)	Resistor	10k / 20k / 40k / 100k / 680Ω	0805 SMD	31	Biasing, filtering, limiting
C1–C7	Capacitor	100nF / 1μF / 10μF	0805 / 1206	7	Decoupling, filtering
J1–J4	Terminal block	Phoenix 5.08mm pitch	Through-hole	4	External connections

9. Schematic Design



10. PCB Design Details

10.1 Layer Stack-up

Layer	Usage
Front Copper (F.Cu)	All signal routing, 5 V and 12 V power traces
Back Copper (B.Cu)	Solid ground plane — uninterrupted copper pour
Front Silkscreen	Component references, polarity markers, connector labels
Front/Back Solder Mask	Solder mask to prevent bridging

10.2 Trace Widths

Net Type	Trace Width	Rationale
Signal traces	0.25 mm	Low current signal routing
5 V distribution	0.5 mm	Multiple load currents
12 V input / regulator	0.8 – 1.0 mm	Higher current carrying requirement

10.3 Component Placement

The board is organised into functional blocks for clarity and signal flow optimisation:

Board Region	Block
Top-left	Power supply — J1, D1, U1, decoupling capacitors, power LED
Upper-centre	Temperature channel — J2, U2A, U3A/B, Q1, Q2, LEDs
Upper-right	Light channel — J3 (bottom), U2B, light LEDs
Lower-centre	Pressure channel — J4, U3C/D, Q3, Q4, LEDs
Right edge	All LED indicators grouped for easy observation

10.4 Grounding Strategy

A solid copper ground plane on the bottom layer provides a low-impedance return path for all circuits, reduces noise coupling between channels, and simplifies routing by eliminating the need to route individual GND traces.

10.5 Design Verification

Check	Result
Electrical Rules Check (ERC)	0 Errors, 0 Warnings
Design Rules Check (DRC)	0 Errors, 0 Warnings
Gerber file generation	Complete — all 9 layers exported
Drill file	Generated (.drl format)

11. PCB Renders

11.1 3D Top View

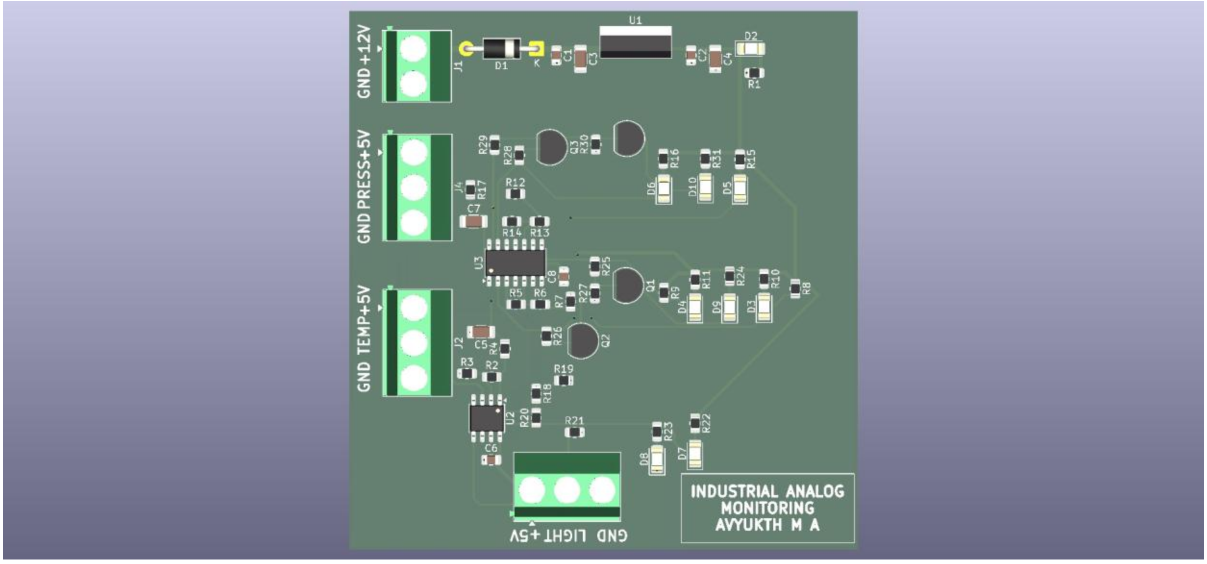


Figure 1 — Industrial Analog Monitoring Board (KiCad 3D Render, Top View)

11.2 PCB Layout (Top + Bottom Overlay)

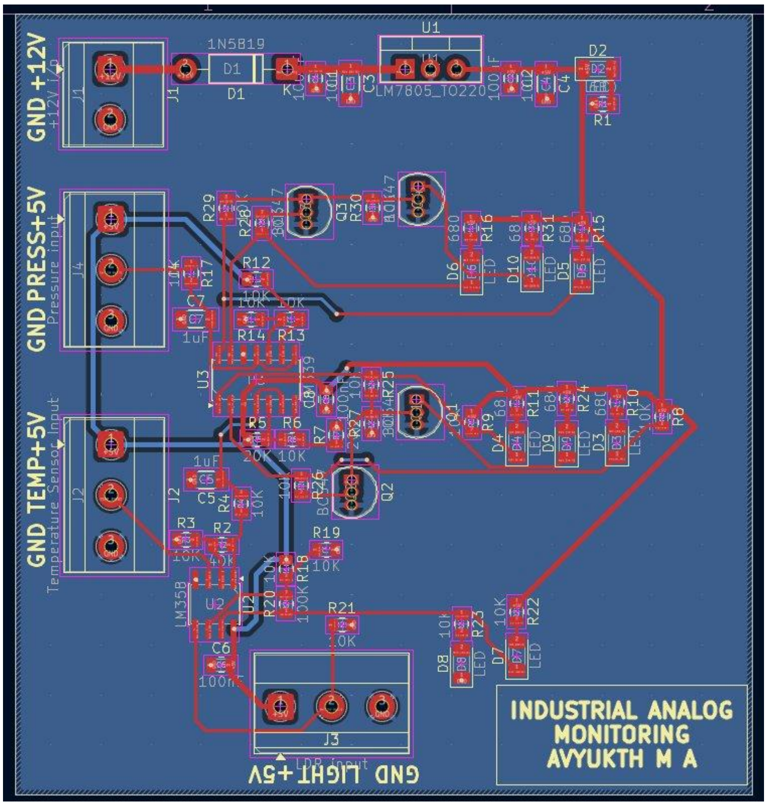


Figure 2 — PCB Layout showing front copper (red), back ground plane (blue), and silkscreen

12. Internship Learning Outcomes

This project served as a practical integration of the Analog Electronics topics covered during Weeks 1–2 of the internship. The following competencies were applied directly in the design:

Skill Area	Application in This Project
Op-amp gain design	Non-inverting amplifier (gain = 5) for LM35 signal conditioning
Comparator circuit design	Dual-threshold detection in temperature and pressure channels
Schmitt trigger / hysteresis	Light channel — noise-immune on/off switching with 100 kΩ feedback
Signal filtering	RC low-pass filters (10 kΩ / 1 μF) on all sensor inputs
Voltage reference generation	Resistor dividers for 2 V, 3 V, 1.67 V, 3.33 V, and 2.5 V references
Transistor switching logic	BC547 AND logic for zone-specific LED control
PCB design workflow	KiCad schematic → layout → DRC → Gerber — complete flow executed
Mixed-technology PCB	SMD components alongside through-hole connectors and transistors
Ground plane design	Solid B.Cu copper pour for noise reduction and clean return paths

13. Conclusion

The Industrial Analog Condition Monitoring Board demonstrates that industrial-grade multi-parameter monitoring can be achieved entirely with analog circuitry — without a microcontroller. The design integrates operational amplifiers, comparators, transistor logic, and signal conditioning techniques into a single, cohesive PCB.

The project successfully passed all ERC and DRC checks in KiCad 10, and a complete Gerber file set has been generated, making the board ready for prototype manufacturing. The design serves as a practical demonstration of the Analog Electronics concepts covered during the Yuken India Ltd. internship and reflects a solid understanding of real-world circuit design practice.

Possible future enhancements to this design include:

- Adding a 4–20 mA current loop input stage for industrial sensor compatibility
- Integrating a buzzer driver for audible alerts at critical thresholds
- Designing a Rev B with adjustable threshold potentiometers on-board
- Extending the light channel to a multi-level LDR comparator arrangement

14. References

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